Hall Ticket Number:

MR-21

Code: B414CS5S3

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2024/January 2025

Computer Organization and Architecture

(Computer Science and Engineering)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	List any two arithmetic micro-operations.	2	1	1	1
1. b)	What are different types of buses used in computers? Why is the address bus unidirectional?	2	2	2	2
1. c)	What are the advantages of microprogrammed control unit?	2	2	2	2
1. d)	What is the purpose of Program Control instructions in a processor?	2	3	2	3
1. e)	Identify the two types of data types used in computer arithmetic.	2	1	1	1
1. f)	Define fixed-point representation used in data representation.	2	2	2	2
1. g)	What is a cache memory? What is its use?	2	1	2	1
1. h)	What are advantages of DMA control?	2	3	3	1
1. i)	What is vector processing, and how does it improve computational performance?	2	2	1	2
1. j)	List and explain the key characteristics of a multiprocessor system.	2	2	2	2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the basic operational concepts of a computer with the help of a suitable diagram.	5	2	1	1,2
2. b)	Explain the bus structure in detail with a neat diagram. Discuss its types and significance in the context of computer architecture.	5	2	2	1,2,3
	OR				
2. c)	Draw the schematic diagram of an Arithmetic Logic Shift Unit (ALU) and explain its working.	5	3	4	1,2,4
2. d)	What is an Instruction Cycle? Illustrate the flowchart diagram of the Instruction Cycle and explain its phases briefly.	5	2	3	1,3
3. a)	Compare the hard-wired control unit and micro programmed control unit	5	4	1	2,4
3. b)	Explain address sequencing process of a microprogrammed control unit with a suitable block diagram?	5	2	2	2,4
	OR				
3. c)	Explain the different types of data transfer and data manipulation instructions in computer with suitable examples.	5	2	3	1,2
3. d)	Distinguish between direct and indirect address instructions. Also, explain the various addressing modes with examples.	5	4	3	2,3
4. a)	Explain the concept of floating-point representation and how it differs from fixed-point representation. Discuss its advantages and disadvantages.	5	2	2	1,3
4. b)	Explain how complements are used in binary arithmetic for subtraction, with suitable example.	5	3	2	1,4
	OR				

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4. c)	Draw and explain a flowchart for a multiplication algorithm.	5	3	1	1
4. d)	Describe the decimal arithmetic operations used in computers. Explain the role of a decimal arithmetic unit in performing these operations effectively in digital systems.	5	4	3	2
5. a)	Explain the need for input-output interface and discuss asynchronous data transfer in detail.	5	2	3	2,4
5. b)	Illustrate interrupt priority schemes with a neat diagram. Explain how these schemes are utilized in managing interrupts.	5	4	4	1,2
	OR				
5. c)	Explain the following Auxiliary memory devices with suitable diagrams: i. Magnetic Disks ii. Magnetic Tape.	5	2	3	4
5. d)	Draw and explain the memory hierarchy in a computer system with necessary block diagrams.	5	3	2	5
6. a)	Differentiate CISC and RISC computer characteristics?	5	2	1	1,2
6. b)	List various parallel processing challenges. Draw the block diagram of 5 stage pipeline system.	5	3	3	1,3
	OR				
6. c)	Explain inter processor communication and synchronization in a shared multiprocessor environment.	5	2	4	2
6. d)	Explain the implementation of Arithmetic and Instructional pipeline.	5	3	2	4

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome